

# Groups & Representations

## Exercise set 1

February 27, 2026

14:30-17:30

### Learning outcomes of this session

- Understand conserved quantities in quantum mechanics.
- Being able to check the axioms of a group for examples.
- Being familiar with some of the most encountered groups that appear in physics.
- Familiarity with the Galilei group.
- Being able to prove basic properties of groups.
- Being able to apply these basic properties in other contexts (applied to specific groups, in other proofs,...).

## 1 Symmetries in classical and quantum systems - Exercises to Chapter 1

### Symmetries in physics

Symmetries form the cornerstone of our temporary understanding of physics. A symmetry is an action on a physical system that leaves the dynamics, ie. the equations of motion, invariant. Symmetries appear all over the place in physics, most notably in quantum physics and relativity theory, but also in classical physics symmetries are often a tool to gain insight in the dynamics of the system and sometimes even provide a way to solve the system completely.

- 1.1. ★★ Consider a quantum mechanical system that evolves in time according to some time-independent Hamiltonian  $\hat{H}$ , show that the expectation value of every time-independent operator that commutes with  $\hat{H}$  is conserved in time.

## 2 The definition of a group - Exercises to Section 2.1, 2.3

### The definition of a group

A *group*  $G$  is a set  $\{g_1, g_2, \dots\}$  endowed with a group multiplication  $G \times G \mapsto G : (g_1, g_2) \rightarrow g_1 \cdot g_2 =: g_{12}$  that satisfies following properties.

- There is a unit element for the multiplication, usually denoted by  $e$  or  $1$  (or  $0$  in case of abelian groups).
- The multiplication is associative.
- For every element  $g \in G$  there exists an inverse element  $g^{-1}$  such that  $gg^{-1} = g^{-1}g = e$ .

If the group multiplication is commutative, we call the group a *commutative* or *abelian* group. In that case the group multiplication is often denoted by  $+$  and the unit as  $0$ . If the group is finite, the number of group elements is called the *cardinality* or *order* of the group.

**Remark:** In the theory lectures we restricted to finite groups, here we also consider infinite groups.

2.1.  $\boxed{\star}$  Which of the following examples are (possibly infinite) groups? What is the order of the groups you find? Are they abelian or not?

- The integers  $\mathbb{Z}_N = \{0, 1, \dots, N-1\}$  with group multiplication addition modulo  $N$ .
- The set of natural numbers  $\mathbb{N}$  with group multiplication addition.
- The integers  $\mathbb{Z}$  with group multiplication addition.
- The even integers  $2\mathbb{Z}$  with group multiplication addition.
- The permutations of a set of  $N$  elements with group multiplication the composition of functions.
- $\text{SO}_n := \{M \in \mathbb{R}^{n \times n} | \det M = 1, M^{-1} = M^\top\}$  with group multiplication the ordinary matrix multiplication.

2.2.  $\boxed{\star\star}$  Recall from a previous course on classical mechanics that an (Newtonian) inertial coordinate frame is one in which Newton's first law holds, i.e. a mass on which no net force acts, will keep on moving with constant velocity. An example of a coordinate frame which is not inertial is for example a rotating frame in which particles are subject to fictitious forces such as the Coriolis force.

A key question in (classical) physics is under which coordinate transformations the laws of physics are invariant. In other words, which coordinate transformations transform one inertial frame into another. These coordinate transformations are exactly the *Galilei transformations*. They correspond to:

- Translations in space and time:  $(\vec{x}, t)^\top \rightarrow (\vec{x} + \vec{r}, t + s)^\top$ . Here,  $\vec{r}$  is a constant displacement vector and  $s$  is some time interval.
- Spatial rotations and reflections:  $\vec{x}^\top \rightarrow O\vec{x}^\top$ , where  $O \in \text{O}_3 := \{M \in \mathbb{R}^{3 \times 3} | M^{-1} = M^\top\}$  is the group of orthogonal 3 by 3 matrices, describing rotations and reflections in 3-dimensional space.

- Uniform motions or *Galileian boosts*:  $(\vec{x}, t)^\top \rightarrow (\vec{x} + \vec{v}t, t)^\top$ . In this notation  $\vec{v}$  denotes a constant velocity vector.

The most general Galilei transformation will be denoted by  $\mathcal{G}(O, \vec{v}, \vec{r}, s)$  and acts via

$$\mathcal{G}(O, \vec{v}, \vec{r}, s) : \begin{pmatrix} \vec{x} \\ t \end{pmatrix} \rightarrow \begin{pmatrix} O & \vec{v} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \vec{x} \\ t \end{pmatrix} + \begin{pmatrix} \vec{r} \\ s \end{pmatrix}. \quad (1)$$

- (a) Compute the composition of two Galilei transformations

$$\mathcal{G}(O_1, \vec{v}_1, \vec{r}_1, s_1) \cdot \mathcal{G}(O_2, \vec{v}_2, \vec{r}_2, s_2). \quad (2)$$

- (b) Show that every Galilei transformation has an inverse given by

$$\mathcal{G}(O, \vec{v}, \vec{r}, s)^{-1} = \mathcal{G}(O^{-1}, -O^{-1}\vec{v}, O^{-1}(s\vec{v} - \vec{r}), -s). \quad (3)$$

- (c) Argue that the Galilei transformations form a group.

- (d) Show that the transformations  $\left\{ \mathcal{G}(\mathbb{1}, \vec{0}, \vec{r}, s) \mid \vec{r} \in \mathbb{R}^3, s \in \mathbb{R} \right\}$  form a normal subgroup of the Galilei group.

- 2.3.  $\boxed{\star}$  Consider a square and find all its symmetries. You should find 8 symmetries. Write down the group structure of the symmetry group, which is  $D_4$ , in two different ways: using generators and relations, and by writing down the Cayley table.

- 2.4.  $\boxed{\star}$  *Rearrangement theorem*: Given some finite group  $G$ , show that every row and column of the Cayley table contains every group element exactly once.